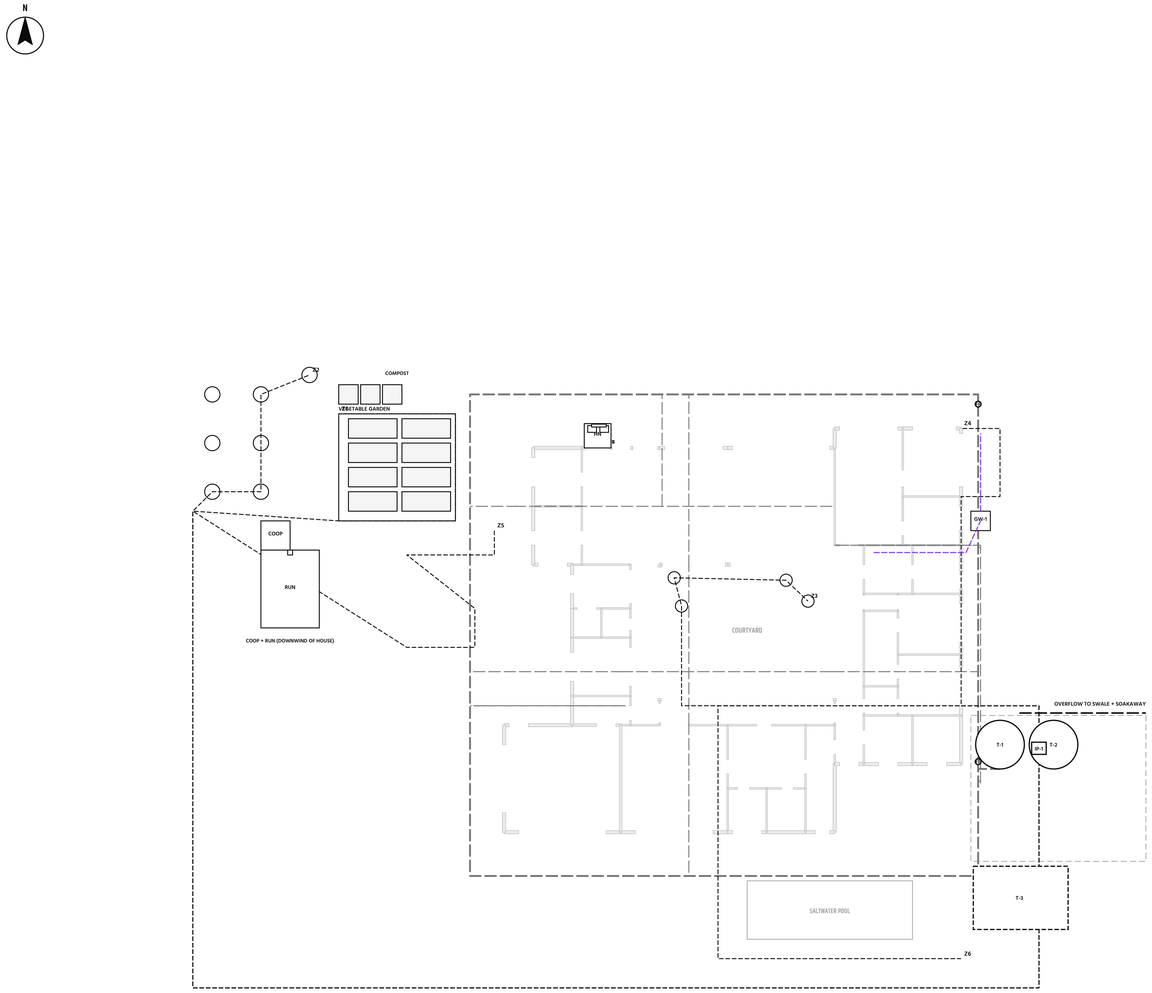

1 **RAINWATER, IRRIGATION + KITCHEN GARDEN PLAN**

SCALE: 1/16" = 1'-0"

CISTERN SIZING

Roof catchment 815 m² (all planes incl. eaves); about 35% (living, master, east) piped to the garden cisterns, the rest to swales. Rainfall about 1,207 mm/yr, ~88% May-Oct (monsoon). Runoff factor 0.8 (metal roof, first flush). Dry-season (Nov-Apr) drip demand 142 m³ for the vegetable garden (49 m²), 4 maples, 7 orchard trees and ~120 m² planting.

MONTH	INFLOW M³	DEMAND M³	SUPPLIED M³	STORED M³
Oct	27.4	8	8	72.0
Nov	9.1	18	18	62.0
Dec	2.7	22	22	42.7
Jan	1.8	24	24	20.6
Feb	2.7	24	23.3	0.0
Mar	5.7	27	5.7	0.0
Apr	13.7	27	13.7	0.0
May	36.5	10	10	26.5
Jun	34.2	4	4	56.7
Jul	38.8	4	4	76.0
Aug	50.2	3	3	77.0
Sep	52.5	3	3	77.0

OPTIONS

TOTAL STORAGE	DRY-SEASON COVERAGE	ROUGH THB (INCL. GUTTERS, FF, PUMP, DRIP)
38 m³	46%	415,000
60 m³	61%	570,000
80 m³	75%	710,000
120 m³	100%	990,000

Selected: 80 m³ = T-1 + T-2 (2 x 19 m³, utility yard) + buried T-3 (42 m³ crate tank). Covers about 75% of dry-season irrigation; the balance (Jan-Apr) is topped up from the well through an air gap, controlled by tank level + soil moisture.

LEGEND

- Drip zone lateral (Z1-Z6)
- Rainwater collector + leaders
- Buried cistern
- Tree / maple (bubblers)

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
1/16" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

RAINWATER NOTES

Gutters on every low eave with leaf screens; first-flush diverters (~1 mm/m²) at both ring inlets.

Garden water is non-potable: purple pipe, labels, no cross-connection with house supply.

Overflow to a planted swale + soakaway down-slope, away from the drain field.

SHEET TITLE

Rainwater, Irrigation + Kitchen Garden Plan

SHEET

P-102